Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions

Made with GAMMA

Project Overview

Dataset Size

3,900 purchases across 18 columns with comprehensive customer and transaction data

Objective

Uncover spending patterns, customer segments, product preferences, and subscription behavior

Goal

Guide strategic business decisions with data-driven insights

This comprehensive analysis combines Python for data preparation, SQL for structured business questions, and Power BI for interactive visualization. The dataset includes customer demographics, purchase details, and shopping behavior metrics.

Made with GAMMA

Data Preparation & Cleaning

Initial Exploration

Imported dataset using `pandas` and conducted initial exploration with `df.info()` and `.describe()` for summary statistics.

Missing Data Handling

37 missing values in Review Rating column were imputed using median ratings by product category.

Feature Engineering

- Created `age_group` by binning customer ages
- Created `purchase_frequency_days` from purchase data
- Standardized columns to snake case

Database Integration

Connected to PostgreSQL and loaded cleaned DataFrame for SQL analysis.

Made with GAMMA

Revenue by Gender

$157,890

Male Customers

Generated 68% of total revenue

$75,191

Female Customers

Generated 32% of total revenue

2.1x

Revenue Ratio

Male customers spent 2.1x more than female customers

Male customers generated significantly higher revenue compared to female customers, representing 68% of total sales. This insight suggests potential opportunities for targeted marketing campaigns to increase female customer spending.

Made with GAMMA

Product Performance Insights

Top Product: Jewelry

171 orders in Accessories category with strong revenue contribution

Top Product: Blouse

171 orders in Clothing category, leading sales in apparel

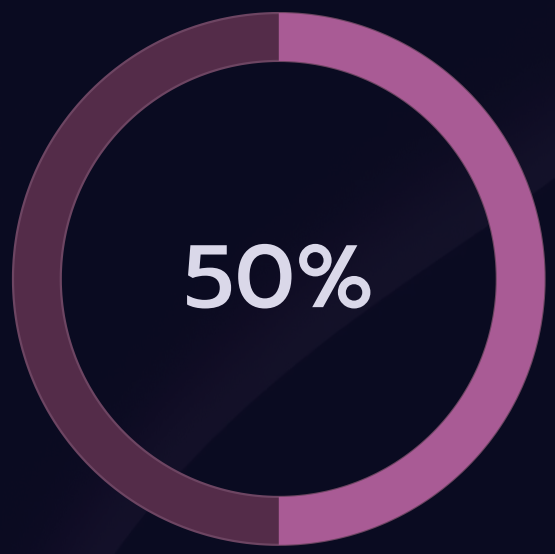
Top Product: Pants

171 orders in Clothing category, matching blouse popularity

Three products tied for highest order volume at 171 purchases each: Jewelry (Accessories), Blouse (Clothing), and Pants (Clothing). These products represent strong performers that could benefit from promotional focus.

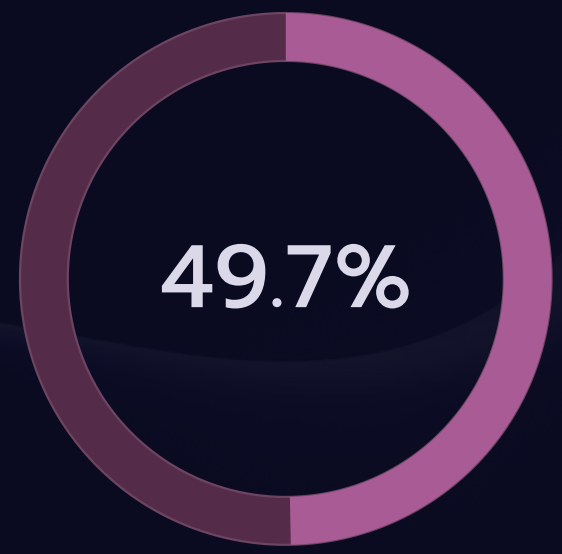
Made with GAMMA

Discount Strategy Analysis



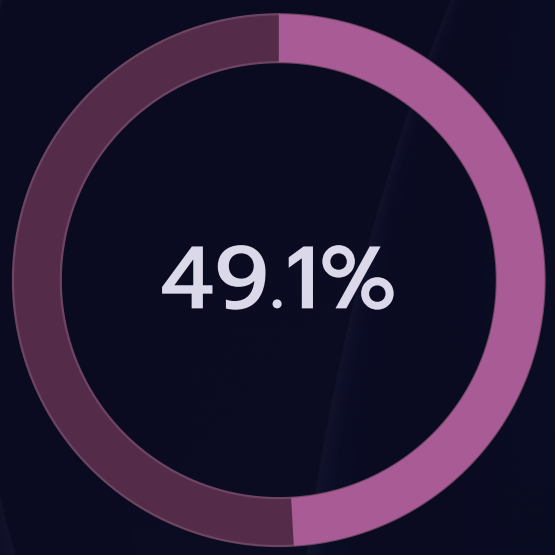
Hats

Most discounted product



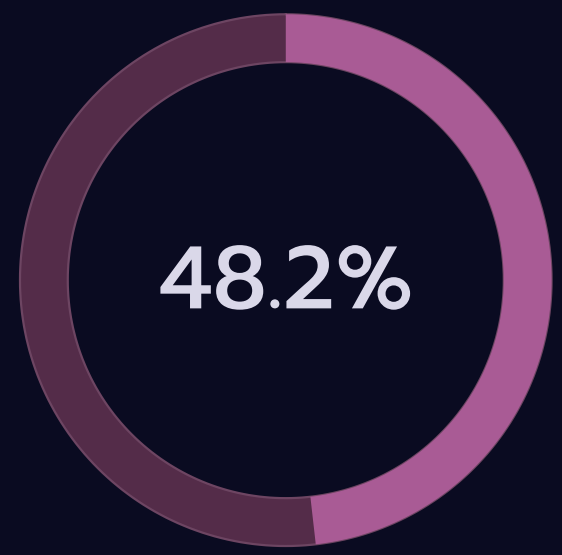
Sneakers

Second highest discount rate



Coats

Third highest discount rate



Sweaters

Fourth highest discount rate

High-value items like hats, sneakers, coats, and sweaters received the highest discount rates. While discounts drive sales volume, businesses should balance this strategy with margin control to maintain profitability.

Made with GAMMA

Shipping & Subscription Impact

Express Shipping

Average purchase: **$60.48**

Standard shipping: **$58.46**

Express customers spend ~3.5% more per transaction

Subscription Status

Subscribers: **1,053 customers**

Non-subscribers: **2,847 customers**

Non-subscribers generated 73% of revenue despite lower subscription rate

Made with GAMMA

Customer Segmentation

Loyal

3,116 customers with consistent purchase history

Returning

701 customers making second or third purchases

New

83 customers with single purchase

The vast majority of customers (80%) fall into the loyal segment, indicating strong retention. However, only 83 new customers represent a small acquisition base that could benefit from targeted marketing campaigns.

Made with GAMMA

Revenue by Age Group

27%

Young Adult

$62,143 revenue

25%

Middle-aged

$59,197 revenue

24%

Adult

$55,978 revenue

24%

Senior

$55,763 revenue

Revenue distribution across age groups is relatively balanced, with Young Adults leading at 27% of total revenue. This suggests broad appeal across demographics, with opportunities to target specific age groups with tailored messaging.

Made with GAMMA

Strategic Business Recommendations

01

Boost Subscriptions

Promote exclusive benefits and incentives to increase subscription conversion rates

02

Enhance Loyalty Programs

Reward repeat buyers to accelerate movement into loyal customer segment

03

Optimize Discount Policy

Balance sales volume with margin control, especially for high-discount products

04

Highlight Top Products

Feature best-rated and best-selling products in marketing campaigns

05

Target High-Value Segments

Focus marketing on express-shipping users and high-revenue age groups

Made with GAMMA